YASHAS CHANDRA BATHINI *Full-Stack AI Engineer | Python | FastAPI | Microservices | Agentic AI | REST APIs*

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yashas144.github.io/AI-Engineer-Portfolio F-1 OPT | Authorized to Work in the U.S.

PROFILE

Full-Stack AI Engineer with 5 years of experience designing and deploying Agentic AI systems, LLM-powered applications, MLOps pipelines, distributed microservices, and cloud-native platforms. Experienced in autonomous AI agents, Retrieval-Augmented Generation (RAG), real-time analytics, and scalable API architectures. Proven expertise building production-grade AI solutions using Python, Java, FastAPI, React, AWS and Azure. Strong background in event-driven systems, cloud deployments, and high-performance AI apps serving mission-critical business operations.

SKILLS

Agentic AI & LLMs: Transformers, Hugging Face, RAG, Agentic AI, LangChain, MCP, Crew AI, Semantic Kernel, NLP, FAISS, ChromaDB.

Full-Stack Web Development: React, Spring Boot, Microservices, FastAPI, Flask, REST APIs, Webhooks, LLM Integration.

Programming.: Python, Java, OOP, Design Patterns, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Cursor, Copilot

MLOps: MLflow, Experiment Tracking, Model Registry, Model Monitoring, Model Versioning, AI Evaluation

Cloud & DevOps: Azure, AWS, Docker, Git, GitHub Actions, CI/CD pipelines.

Data Engineering: Azure AI ecosystem, Azure Databricks, ETL Data Pipelines, Event-Driven Architecture, Tableau, MySQL, PostgreSQL.

WORK EXPERIENCE

Full-Stack AI Engineer 06/2021 – 07/2026

Sysarket Hyderabad, India

AfterHours AI - AI Voice Booking & Agent Orchestration Platform

- Designed and deployed a production-grade conversational AI orchestration system, integrating an AI voice agent with FastAPI services and external APIs to automate appointment workflows with zero manual data entry.
- Engineered asynchronous, event-driven processing and WebSocket communication, supporting 10,000+ daily events with sub-second real-time updates to a React dashboard.
- Built resilient distributed workflow controls including event-ID idempotency, retry-safe API integrations, and duplicate-event prevention across Google Calendar and Sheets APIs.
- Validated webhook payloads with Pydantic at the API boundary to catch malformed data before processing.

Auravance – Industrial Predictive Intelligence Platform

- Built a full-stack AI platform (React, FastAPI) for real-time equipment monitoring and failure prediction using XGBoost and LSTM, improving detection accuracy by 30%.
- Engineered scalable event-driven IoT ingestion pipelines using MQTT and Python for high-frequency sensor telemetry with low latency and high reliability.
- Designed scalable ETL and event-driven MQTT pipelines processing 1M+ sensor records/day, enabling predictive maintenance, anomaly detection, and RUL insights with sub-minute latency.
- Built a RAG-based root cause analysis system over equipment manuals to generate explainable diagnostics and maintenance recommendations.

AI-Powered Music Intelligence Platform

- Designed and deployed an AI-powered recommendation platform using Hybrid RAG (BM25 + vector embeddings) across 1M+ songs for context-aware retrieval based on mood, genre and user intent.
- Architected a scalable RAG pipeline incorporating embedding generation, vector search, hybrid retrieval and reranking, improving recommendation relevance and reducing retrieval noise.
- Engineered real-time AI features, including real-time mood detection and seamless playback.
- Deployed a cloud-native full-stack system (React + FastAPI) on AWS and Hugging Face with CI/ CD pipelines, ensuring reliable and scalable delivery.

FIFA Analytics & Prediction System

- Built an ML-driven sports analytics platform with production APIs and interactive dashboards for match prediction.
- Improved model accuracy by 18% using feature engineering, hyperparameter tuning, and MLflow tracking.
- Implemented model registry, versioning, and monitoring (performance, confidence, and feature drift).
- Deployed a containerized full-stack application using React (frontend) and Java (backend) with integrated ML services, leveraging Docker and Azure with Azure SQL DB for scalable, production-ready model serving.

Deep Learning-Based Image Synthesis with GAN Architecture | Research Publication

- Developed GAN-based image synthesis models using PyTorch and TensorFlow, improving output quality.
- Contributed to generative AI research by designing GAN architectures for real-world applications.

EDUCATION

Master of Science in Data Science Denton, Texas

University of North Texas, CGPA: 4.0 / 4.0

Bachelor of Technology in Information Technology Hyderabad, India

Jawaharlal Nehru Technological University Hyderabad, CGPA: 3.7 / 4

CERTIFICATIONS

Generative AI Practitioner - AWS | Generative AI Fundamentals - Databricks | Machine Learning with Big Data - UC San Diego | Introduction to Data Science - Cisco | Career Essentials in Generative AI - Microsoft | Python Essentials - Cisco